

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Winter 2024

Course: B. Tech

Branch: Electrical Engineering /Electrical & Electronics Engg &
Allied

Subject Code & Name: BTES305 Engineering Material Science

Semester: III

Max Marks: 60

Date: 21/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. Objective type questions. (Compulsory Question)

12

1

1 Which of the following factors affects the resistivity of metals?

CO1

1

- A) Crystal structure B) Temperature C) Impurities D) All of the above

2 What is the relationship between conductivity and temperature in metals?

CO2

1

- A) Conductivity increases with temperature B) Conductivity decreases with temperature C) Conductivity is independent of temperature D) Conductivity is zero at all temperatures

3 What is the key principle of quantum free electron theory in conducting materials?

CO3

1

- A) Electron-hole generation B) Free electron behavior in a periodic potential C) Ionic bonding D) Superconductivity in high temperatures

4 Which type of polarization is observed in polar dielectrics?

CO3

1

- A) Electronic polarization B) Ionic polarization C) Orientation polarization D) All of the above

5 The Clausius-Mosotti equation relates which of the following properties?

CO2

1

- A) Dielectric constant and internal field B) Dielectric loss and leakage current C) Temperature and frequency D) Breakdown voltage and polarization

6 The Einstein relation in semiconductors relates which two parameters?

CO4

1

	A) Carrier mobility and carrier density	B) Diffusion constant and mobility	C) Recombination rate and generation rate	D) Electron lifetime and diffusion length		
7	What is the primary application of VLSI technology?				CO5	1
	A) Enhancing semiconductor mobility	B) Increasing integration density of circuits	C) Improving breakdown voltage	D) Reducing carrier recombination		
8	What property differentiates soft magnetic materials from hard magnetic materials?				CO1	1
	A) High coercivity	B) Low permeability	C) Low coercivity and high permeability	D) High resistance		
9	Which of the following materials is ferromagnetic?				CO1	1
	A) Copper	B) Iron	C) Zinc	D) Aluminum		
10	What happens to ferromagnetic materials above the Curie point?				CO3	1
	A) They exhibit diamagnetic properties	B) They lose spontaneous magnetization	C) They retain magnetic properties	D) They become superconductors		
11	Bragg's law is used in which type of material testing?				CO1	1
	A) Ultrasonic testing	B) X-ray diffraction	C) Magnetic domain analysis	D) Non-destructive eddy current testing		
12	Which of the following is an example of non-destructive testing?				CO3	1
	A) Ultrasonic radiography	B) Heat treatment	C) Metal forging	D) Arc welding		
Q. Solve the following.						12
2						
A)	Derive and explain Joule's law. How does it relate to energy dissipation in conducting materials?				CO1	6
B)	Define the following terms				CO1	6
	A) Drift velocity	B) Mobility	C) Collision time			
	D) Mean free path	E) Relaxation time				
Q.3 Solve the following.						12
A)	Explain the structure of crystalline and amorphous crystals using a suitable diagram and example.				CO2	6
B)	What is the definition of polarization in dielectric materials and what is its classification?				CO2	6

Q. Solve Any Two of the following. 12

4

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|--|-----|---|
| A) Explain the Einstein relation in semiconductors and its significance in understanding carrier mobility and diffusion. | CO3 | 6 |
| B) What is Semiconductor? Write its properties in detail. | CO3 | 6 |
| C) Explain intrinsic semiconductor and extrinsic semiconductor with suitable examples | CO3 | 6 |

Q.5 Solve Any Two of the following. 12

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|---|-----|---|
| A) Classify magnetic materials and explain the differences between diamagnetic, paramagnetic, and ferromagnetic materials. | CO4 | 6 |
| B) What is curie temperature? And what are the significance of curie temperature/curie point in magnetic material? | CO4 | 6 |
| C) Explain the properties of magnetically soft and hard materials. Discuss their applications in electrical machines and instruments. | CO4 | 6 |

Q. Solve Any Two of the following. 12

6

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| A) Explain Bragg's law and its significance in X-ray crystallography. How does it relate to the interference of X-rays in a crystal lattice? | CO3 | 6 |
| B) What is radioactivity, and how do radioactive materials differ from non-radioactive ones? | CO3 | 6 |
| C) What is ultra-sonic radiography? explain in detail. | CO3 | 6 |

*** End ***

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